TekFrameworks

Modern AI Models of Text

Embedding, Transformer, Training

Academic
Ph.D. Computer Science (Machine Learning)
Johns Hopkins University – USA, 1995
M. Tech. Computer Science (Knowledge Based Systems), Indian Institute of Technology (IIT), Madras, India
B.E. Computer Science: MNNIT, Allahabad, India

Research
Dozens of Doctoral and Masters' theses
Trained 10,000 AI & data science practitioners
50+ Patents, Publications
Invited talks at Trilateral Commission, Concordia Summit, AAAI, IJCAI, KDD, and CII National Summit

Accolades
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"First Indian to earn a PhD in Machine Learning"
Among the "10 most prominent analytics academicians in India"
"AI Visionary", "Big Data Rockstar"

Instructor Profile

Dr. Sreerama K. Murthy

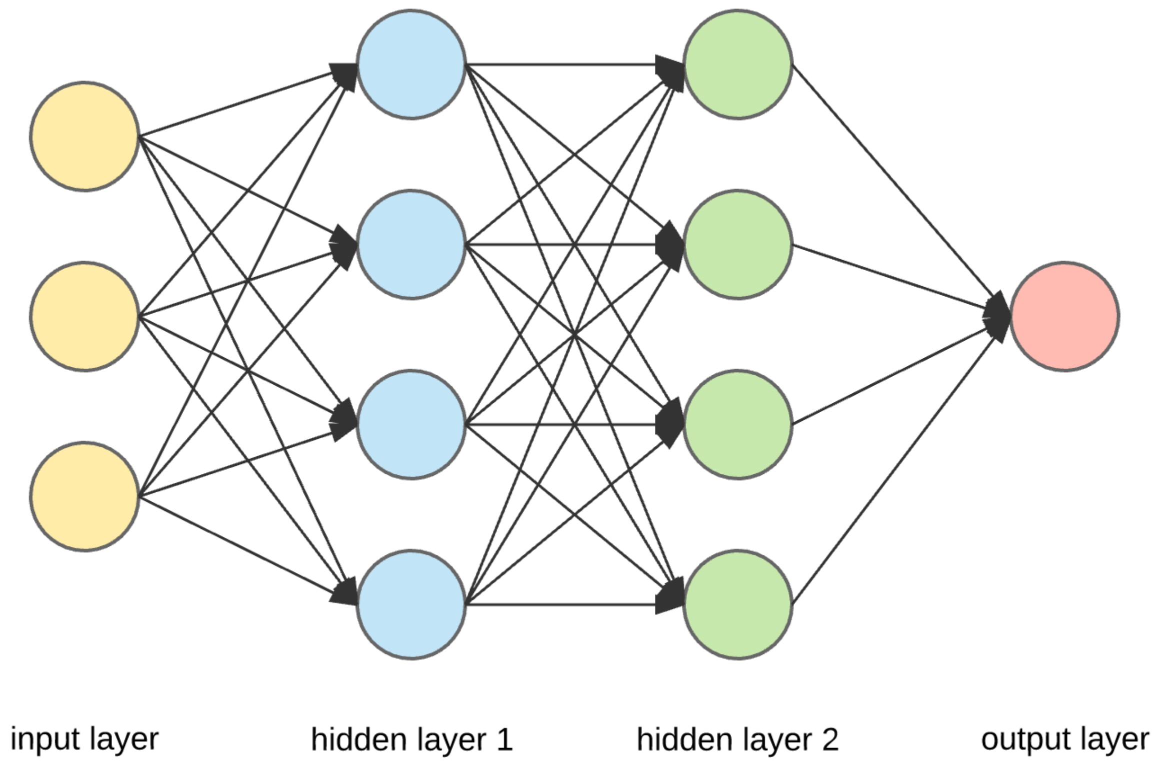
Work Experience
30+ years of experience in AI
Scientist, Serial Entrepreneur
Setup companies in India, US, South America
Brought in millions of dollars of funding
IBM Research, Siemens Corporate Research
Westinghouse, National Center for Software Technology
Professor,: Insofe, upGrad Insofe
Adjunct professor: Case Western U. (USA), Golden Gate U. (USA), Renne School of Business (France)

Sample AI Systems Built
Banking fraud detection, drug side effects discovery, video content extraction, serendipitous ad placement, boosting
revenue efficiency, predicting & controlling staff attrition, agri demand forecasting, document automation, conversational bots, energy loss minimization, image & big text
analytics for the Hubble Space Telescope, deep diagnosis tools for cancer, AI-delivered coaching & vocational skilling, cognitive supply chains,

In the next 90 minutes…

- Assumption for this session: Participants all know what a neural network is. We will refresh rapidly

- Content
  - ✓How do we input Text into any AI model?
  - ✓What is a transformer?
    - ✓ Encoders, Decoders, Encoder-Decoder Combinations
  - ✓How do we train these transformers?



Inputs (Embeddings)

The major shift is from treating text as a bag of words to a semantically meaningful, context dependent artifact.

Tokenizing text

We need to convert text into numbers that neural networks can manipulate.

- First text is automatically converted to bits and bytes (Convert Text to Binary Code (Base2 Character & Word Encoder) - B64Encode) in a computer.

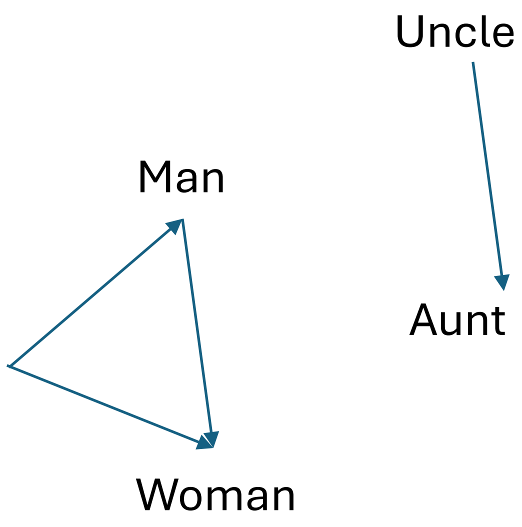
- Greedily combine the most frequent adjacent pairs and proceed to build combinations. This is called tokenization. Token can be a word, word and punctuation, punctuation alone or part of a word. Tokenizers are case, space sensitive.

- Each large model has its own tokenizer. ChatGPT 4.0 uses Tiktokens (Tokenizer - OpenAI API) and it converted English into 100,277 unique tokens.

Vectorizing a tokenizer

- Tokens are arbitrarily assigned integers. We need to convert them to meaningful representations. We can use one-hot encoding. But, this is sparse and not meaningful.

- Google Word2Vec showed how to represent words/tokens as vectors of arbitrarily high dimensions. Each dimension can be thought of as a separate concept (independent)
  - All similar words group in one location
  - Directions capture the relationships
    - Uncle – Man + Woman = Aunt



- Tokens are initialized into high dimensional vectors and learned. According to its own admission, ChatGPT uses a pre-learned 12,288 dimension vector for each token.

Context & Position Embedding

- The context length is how many such tokens can be taken in one prompt (128K for later ChatGPT). We pad where needed if the length is smaller

- Modern day models are parallelized. Which means, we will lose the crucial sequence information during processing. Hence, we add the position information explicitly.

- These position embeddings are added to the token embeddings for complete packaging of information and then layer normalized.

Architecture

Major shift is understanding that conversation is a combination of context, and knowledge and style. Create blocks that address these

How is Conversation Built in LLMs

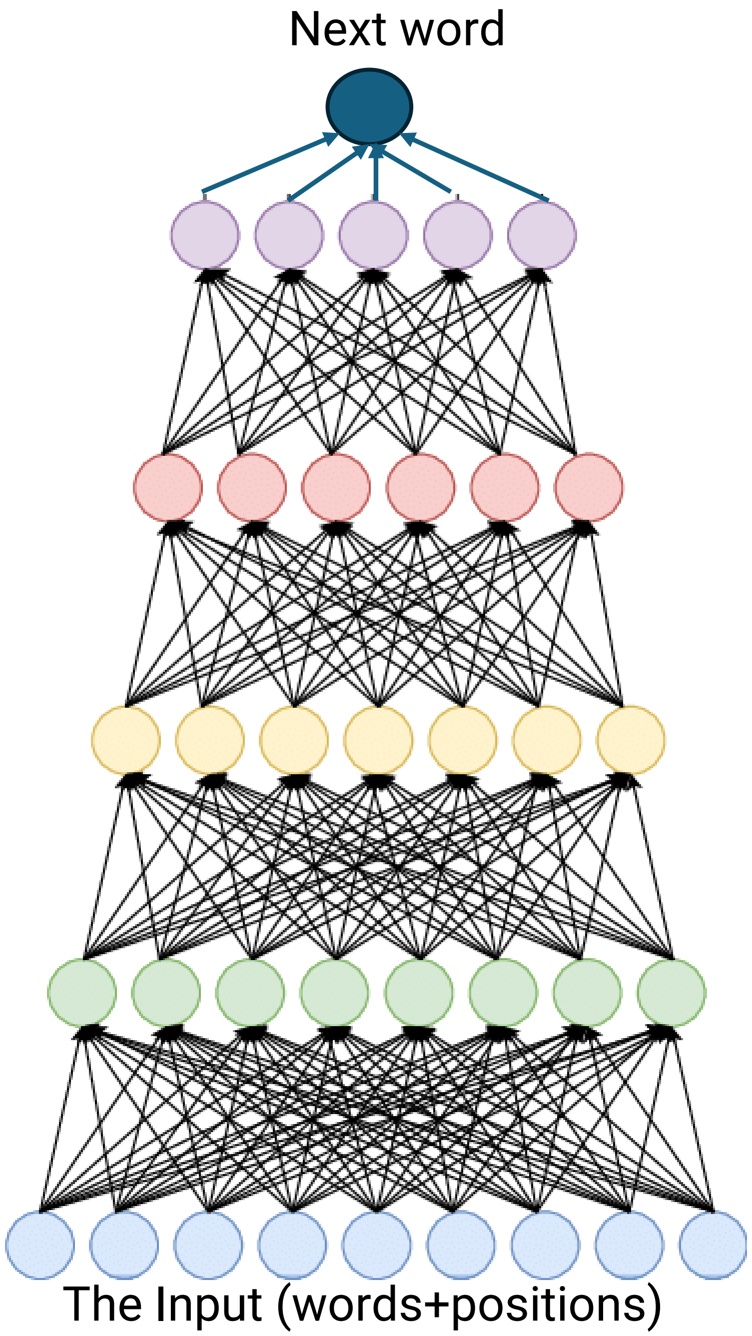
- They start with predicting the next word given a context. This is the unit operation in any text generation. This is called language modelling. What does predicting the next word take?
    - Ability to understand the context
        - This is a glass of orange ________ (n-1 and n-3)
    - Ability to store and use the knowledge
        - Total century count of Kohli in one-days is ________ (n-2, n-4, n-6, n-7) and still this completion needs some look up knowledge

What about a large neural network

- Neural network is great for storing knowledge

- However, context

  - This is a glass of orange ________ (n-1 and n-3)
  - The clouds are in the _________ (n-4)
  - Total century count of Kohli in one-days is ________ (n-2, n-4, n-6, n-7) and still this completion needs some look up knowledge



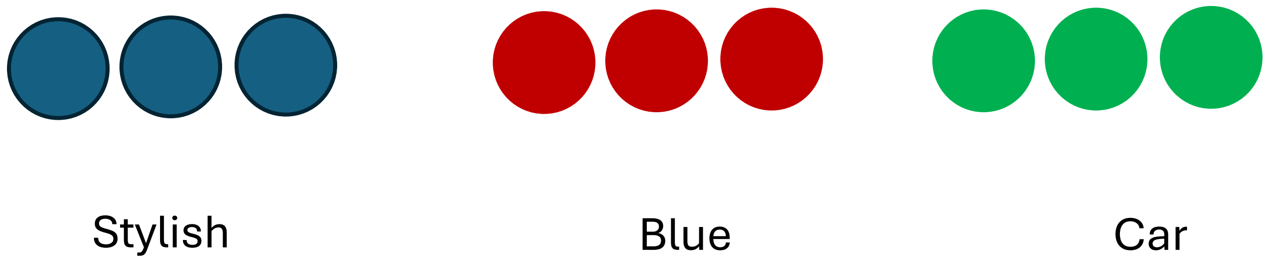
Attention block is the context finder

A separate intervention called “***attention***” handles this problem elegantly. Attention updates the vector of each word based on the context.

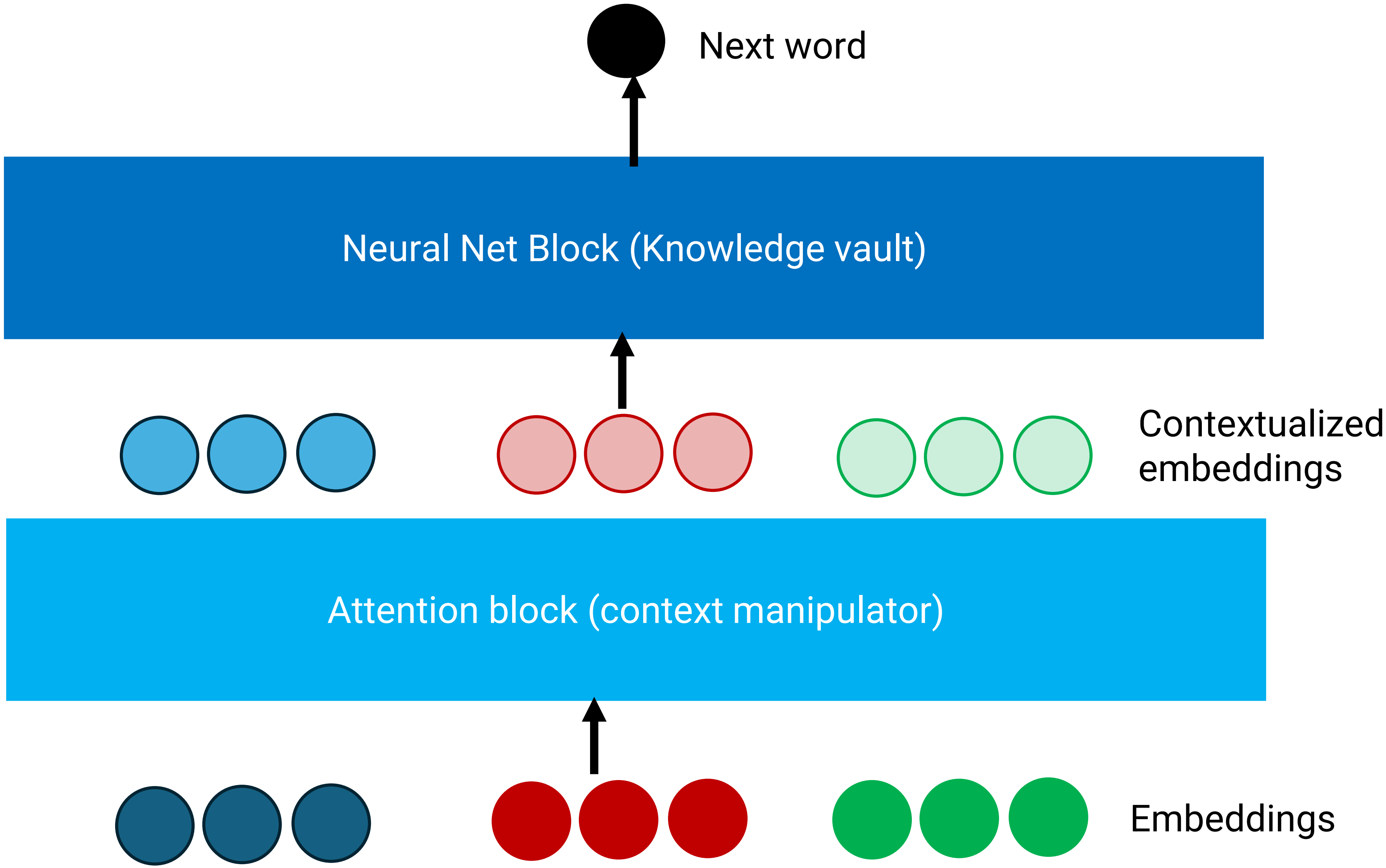
Every word is updated by every other word.

From the embeddings, we initialize query, key and value vectors. Then, $(q.k^T)$. is used to update token vector

After analysing a lot of data, the model learns to update embeddings such that appropriate words influence the word a lot and irrelevant words influence it a little.



A Transformer's Architecture



FFN typically stores knowledge

Attention has the ability to manipulate the context.

This combination is called a transformer.

Attention is all you need

An apt title could be "Attention is 30% of all you need!!!" as NNs have 70% weights

Encoder only: Good for classification (BERT)
Decoder only (without cross attention): Generation (GPT)
Encoder + Decoder: Translation

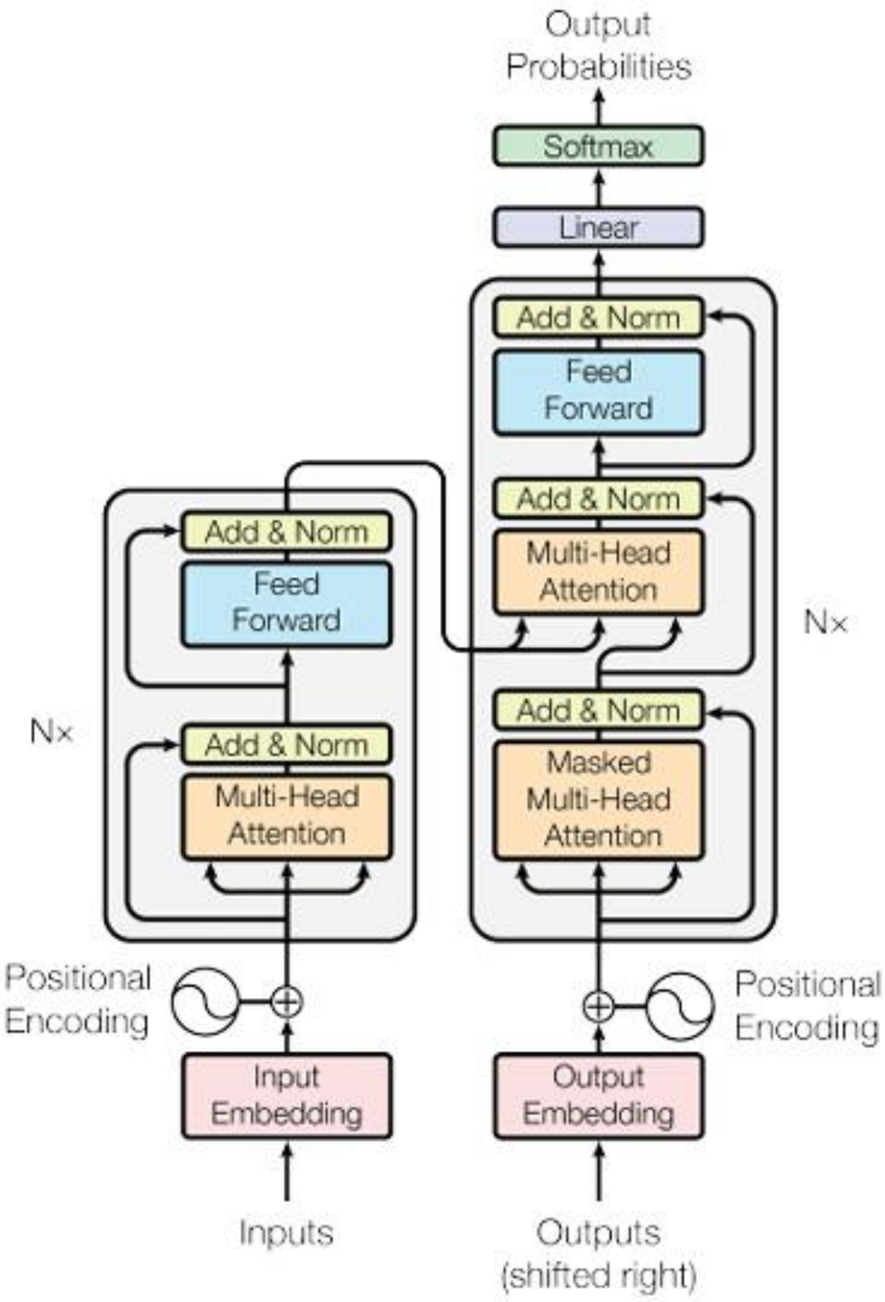
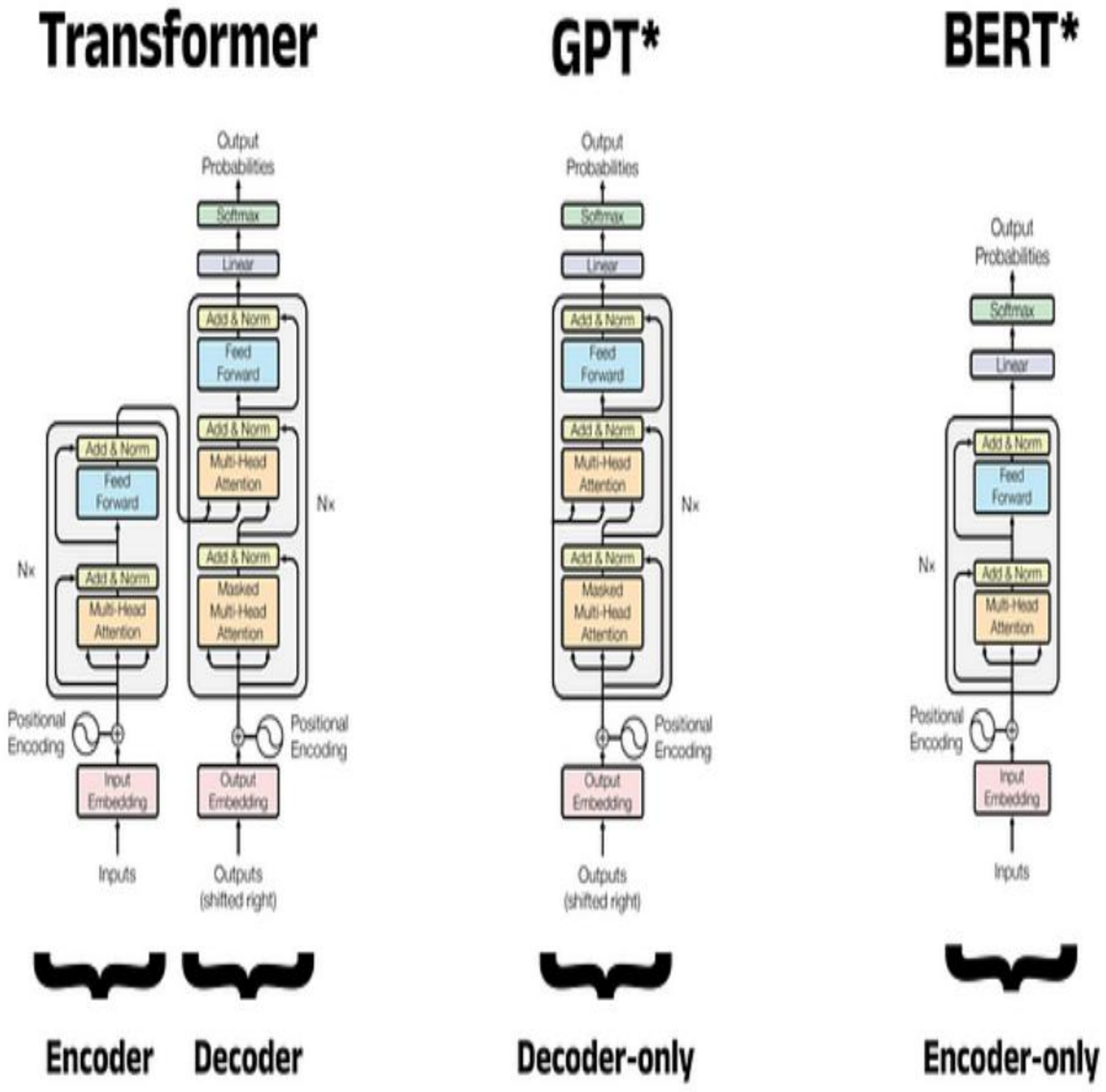


Figure 1: The Transformer - model architecture.

Feature	GPT	BERT	T5
Main Use	Text generation	Text understanding	Multi-purpose NLP tasks
How it reads text	Left to right (unidirectional)	Both directions (bidirectional)	Both + decoder
Best for…	Chatbots, content creation	Search engines, classification	Summarization, translation, Q&A
Biggest Weakness	Can be inaccurate at times	Can't generate text	Requires a lot of computing power



https://automotivevisions.wordpress.com/2025/03/21/comparing-large-language-models-gpt-vs-bert-vs-t5/

Gen AI (specifically LLM) is a humongous DL

TokenIDs (1×4096)

↓

Embedding: (4096×50257) · (50257×12288) → (4096×12288)

↓

QKV projection: (4096×12288) · (12288×36864) → (4096×36864)

↓

Split → Q(4096×12288), K(4096×12288), V(4096×12288)

↓

Attention: (4096×12288)·(12288×4096) → (4096×4096)

↓

Weights × V: (4096×4096) · (4096×12288) → (4096×12288)

↓

ChatGPT 3 (175 Billion Parameters)

Output proj: (4096×12288) · (12288×12288) → (4096×12288)

↓

FFN expand: (4096×12288) · (12288×49152) → (4096×49152)

↓

FFN contract: (4096×49152) · (49152×12288) → (4096×12288)

↓

Final row: (1×12288)

↓

Vocab projection: (1×12288) · (12288×50257) → (1×50257)

↓

Softmax → next-token distribution

Training

Supervised, Unsupervised and Reinforcement learning

A training pipeline that combines context, knowledge and style

Large Data sets for both

- Well curated datasets (fineweb)
  - HuggingFaceFW/fineweb · Datasets at Hugging Face
- Predict the next token given some previous tokens
  - This is called base model

Supervised learning

- What does generating a whole conversation take (Ability to mimic human style)
- Make people write prompts and answers and train

Limitations of Pre-Training and SFT

- Pre-training and SFT do not provide an effective way to steer a model away from bad outputs.
  - The pre-training dataset contains some data that is unkind, so a model trained with next-token-prediction may sometimes be unkind to the user. How do we explicitly tell the model to be kind to the user?
- They are not also appropriate for specifying abstract goals.
  - E.g. “Follow the user’s instructions helpfully and safely.”
  - E.g. “Refuse a user’s command if it is unethical or dangerous.”
- SFT lacks explicit negative signals. When a behavior is absent from the training data, the model gradually infers its undesirability through omission. This indirect signal is weak and inefficient.

Phases of Training: Reinforcement Learning

- Instead of copying examples, think of it like training a puppy: you reward good behavior (like sitting on command) and discourage bad ones. However, reward is sometimes difficult to specify as a formal function in conversations.

- RL as a field worked well for games, where reward was winning.

- RL with human feedback (RLHF) uses human feedback to determine the desired behavior, often by training a *reward model*.

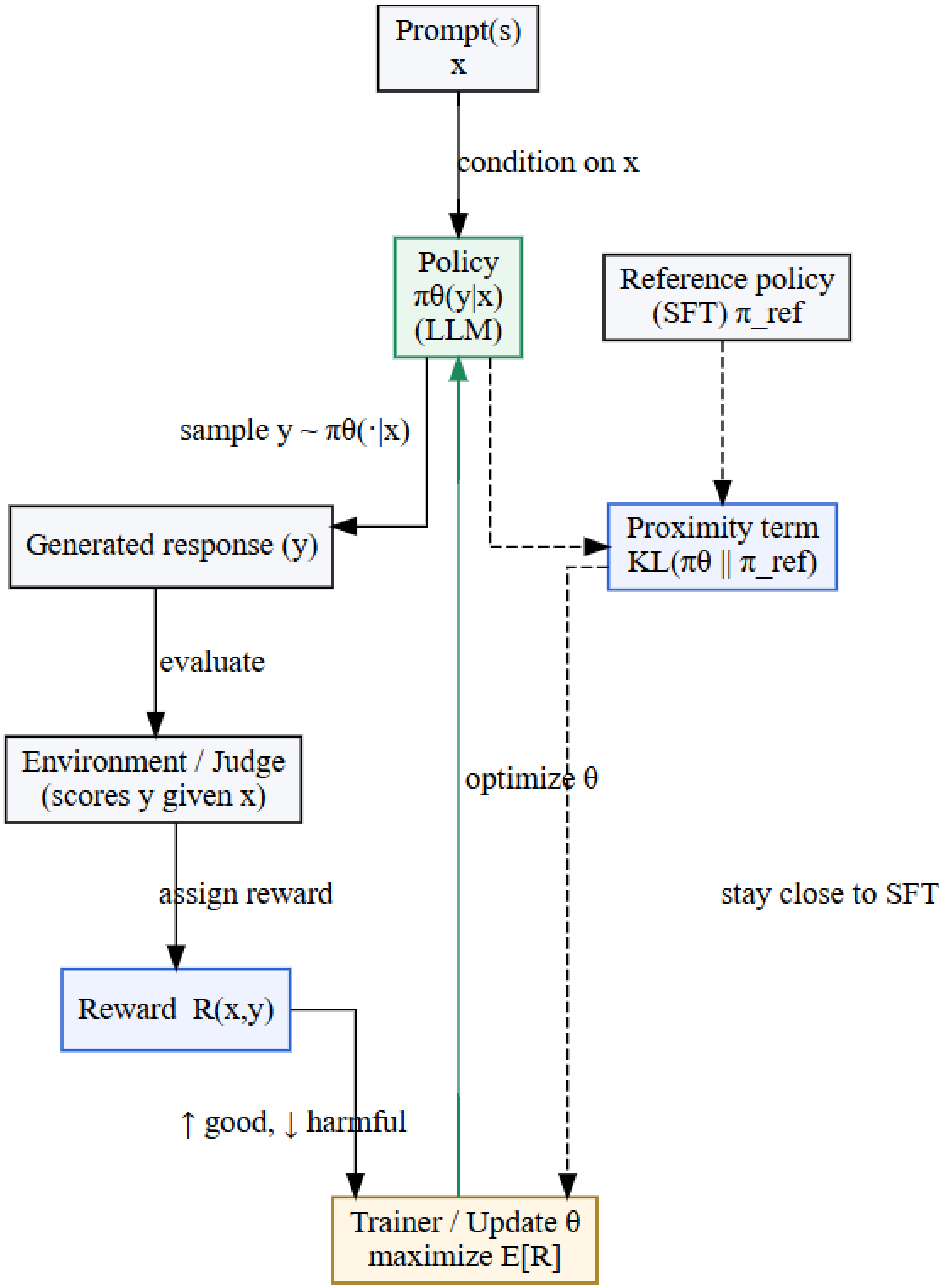
  - P. Christiano, J. Leike, T. B. Brown, M. Martic, S. Legg, and D. Amodei, Deep reinforcement learning from human preferences, *NeurIPS*, 2017.
  - B. Ibarz, J. Leike, T. Pohlen, G. Irving, S. Legg, and D. Amodei, Reward learning from human preferences and demonstrations in Atari, *NeurIPS*, 2018.

Why RL works?

- RL provides explicit negative feedback. Actions that lead to lower rewards are penalized. Although there is no credit assignment and all actions leading to a poor outcome are down weighted without fine-grained attribution, the model nonetheless receives direct signals about which behaviors to avoid.
- Studies show RLHF boosts user satisfaction and cuts toxicity (harmful content) by 50% or more (Askell et al., 2021). Without RL, models might be grammatically perfect but unengaging or risky. RL adds that extra polish, like turning a robot into a friendly companion.

RL process

Verifiable rewards



Models are learning to reason

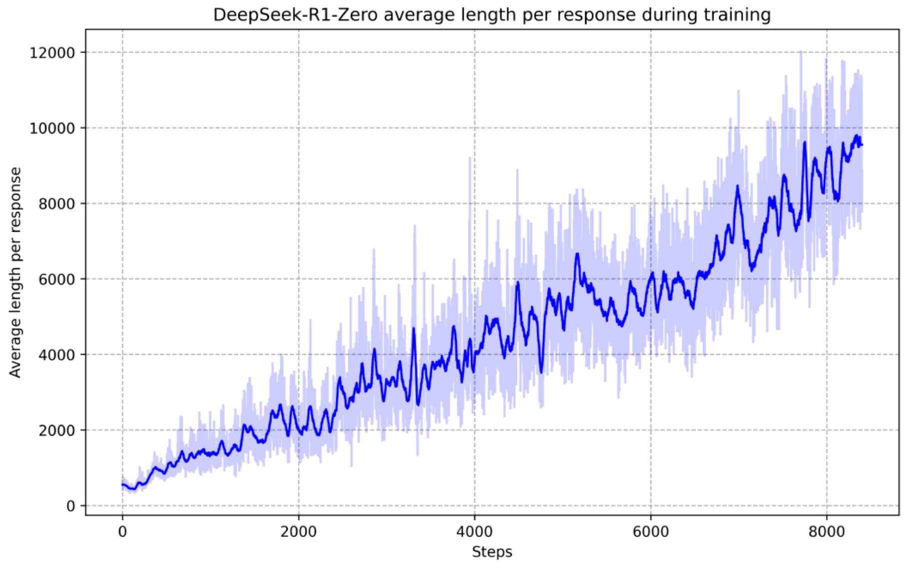


Figure 3 | The average response length of DeepSeek-R1-Zero on the training set during the RL process. DeepSeek-R1-Zero naturally learns to solve reasoning tasks with more thinking time.

Question: If $a > 1$, then the sum of the real solutions of $\sqrt{a - \sqrt{a + x}} = x$ is equal to

Response: <think>

To solve the equation $\sqrt{a - \sqrt{a + x}} = x$, let's start by squaring both $\cdots$

$\left(\sqrt{a - \sqrt{a + x}}\right)^2 = x^2 \implies a - \sqrt{a + x} = x^2.$

Rearrange to isolate the inner square root term:

$(a - x^2)^2 = a + x \implies a^2 - 2ax^2 + (x^2)^2 = a + x \implies x^4 - 2ax^2 - x + (a^2 - a) = 0$

...

Wait, wait. Wait. That's an aha moment I can flag here.

Let's reevaluate this step-by-step to identify if the correct sum can be $\cdots$

We started with the equation:

$\sqrt{a - \sqrt{a + x}} = x$

First, let's square both sides:

$a - \sqrt{a + x} = x^2 \implies \sqrt{a + x} = a - x^2$

Next, I could square both sides again, treating the equation: $\cdots$

...

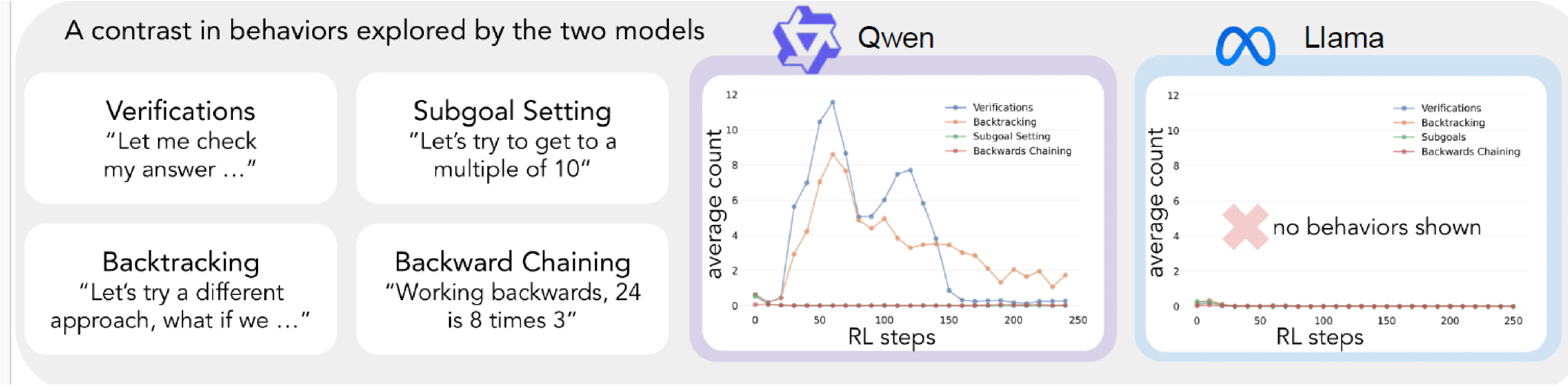
These are learned by just looking at the correct answers

RLVR and Emergence

- “One of the most remarkable aspects of this self-evolution is the emergence of sophisticated behaviors as the test-time computation increases for models trained with RLVR
  - These behaviors are not explicitly programmed but instead emerge as a result of the model’s interaction with the reinforcement learning environment.”
- These worked for Qwen, Deepseek which were trained on RLVR and not Llamma or other models which were not trained!!

RLVR works with Cognitive Behavior

- Qwen and DeepSeek base models exhibit certain key cognitive behaviours for reasoning: verification (systematic error-checking), backtracking (abandoning failing approaches), subgoal setting (decomposing problems into manageable steps), backward chaining (reasoning from desired outcomes to initial inputs).



A Simple Recap

- SFT teaches "what": formats, facts, behaviors.
- Preference training teaches "how well": quality, style, safety.
- RLVR teaches "how to": procedures, strategies, problem-solving.